D2. Candidate LLM Comparison and Selection

Model-selection evidence, criteria, and provisional recommendation.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

Controlled model selection result

Model	Accuracy	Macro-F1	Grounding	Mean latency	Selection meaning
Qwen3 8B	81.25%	0.4296	100.00%	13.66 s	Leading local candidate
Qwen2.5 7B	62.50%	0.4190	100.00%	15.16 s	Stable control model
Llama 3 8B	43.75%	0.2456	56.25%	44.66 s	Not selected

Selection rationale

Qwen3-8B is selected as the leading benchmark candidate because it produced the strongest answer accuracy and retained the evidence-grounding check on the same two-report workload. This is a concrete decision based on completed local runs, not on a model-card claim. [E10]

The choice remains deployment-provisional because the test corpus is small and source-derived. The correct conclusion is that Qwen3-8B is the best observed candidate for this proof of concept, while independent expert labels are required before declaring a production winner. [E10]

Decision statement

Qwen3-8B is the leading local generator candidate: it reached 81.25% accuracy on the common two-report evaluation, ahead of Qwen2.5-7B at 62.50% and Llama-3-8B at 43.75%. Qwen3-0.6B is not selected because its completed run collapsed to the No class. [E3][E10]

This is a benchmark-selection result, not a production-selection claim. Independent, balanced multi-report labels are still required. [E10]

Selection criteria

Criterion	Why it matters	Current decision use
Answer quality	Benchmark decisions must match gold labels	Accuracy and macro-F1; class support shown [E5]
Evidence support	Audit decisions must be traceable	Grounding rate and valid evidence IDs [E5][E8]
Instruction/JSON reliability	Automated scoring needs structured output	Schema-enforced local generation [E8]
Latency and resources	Offline laptop deployment is constrained	Mean latency, CPU/GPU fit and model size [E3][E4]
Governance	Audit data must remain local and reviewable	Offline serving, config, source and output traceability [E4][E8]

Completed evidence

Configuration	Corpus	Observed outcome	Decision
Qwen3 embedding/reranker retrieval	1 report, 12 questions, 10 chunks	Recall@5=1.00; Precision@5=0.2833; MRR=1.00; nDCG@5=1.00	Tied B0 on a small corpus [E2]
Qwen3-0.6B end-to-end	1 report, 12 questions	Accuracy=58.33%; macro-F1=0.2456; all predictions No	Do not select [E3]
Qwen3-8B real-report pipeline	2 reports, 16 questions, 22 chunks	Accuracy=81.25%; macro-F1=0.4296; 13.66 s mean latency	Candidate only; benchmark gate remains blocked [E4][E5]

Justification and limitation

Qwen3-8B can run locally with the documented CPU/GPU allocation, and its two-report run completed with no execution errors. That is integration evidence, not comparative superiority: the labels are source-derived, only two reports are included, and Insufficient Evidence has zero support. [E4][E5]

A final selection should compare Qwen2.5 7B, Qwen3 8B, Qwen3 14B if feasible, and one alternative under frozen prompts, retrieval, questions and hardware. [S4][E8]

Evidence and sources

All numeric claims above are linked to saved metric artifacts [E2]-[E5].

Final model decision ledger

Model	Accuracy / macro-F1	Grounding / latency	Selection outcome
Qwen3 8B	81.25% / 0.4296	100.00% / 13.66 s	Leading POC candidate
Qwen2.5 7B	62.50% / 0.4190	100.00% / 15.16 s	Control baseline
Llama 3 8B	43.75% / 0.2456	56.25% / 44.66 s	Not selected
Qwen3 0.6B	58.33% / 0.2456	All predictions No	Not selected

How to read the decision

Qwen3-8B wins the observed trade-off: it is the highest-accuracy candidate and is faster than the two comparison models in the logged 16-pair run. The selection is therefore evidence-backed for the current hardware and evaluation configuration, rather than based only on model-family reputation. [E10]

However, the No class has recall of 0.25 (one correct No out of four) and the dataset contains no Insufficient Evidence examples. The final report must state this directly: Qwen3-8B is selected for the POC implementation, while production selection remains pending an independently labelled, balanced evaluation set. [E5][E10]

Repository evidence cited

- [E3] results/qwen3_8b_run/1787660483_900adc72559a/: saved Qwen3-8B single-report run artifacts.
- [E4] results/production_real_reports/1787846335_de0ded86a7db/run_config.json and run.log: two-report production-parser run configuration and execution log.
- [E5] results/production_real_reports/1787846335_de0ded86a7db/{retrieval_metrics,question_level_llm_metrics,sota_faithfulness_metrics,audit_level_scores}.json: scored metrics and audit decisions.
- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.

External technology sources

- [S1] Chen et al. (2024), BGE-M3-Embedding, arXiv:2402.03216, https://arxiv.org/abs/2402.03216.
- [S2] Santhanam et al. (2021), ColBERTv2, arXiv:2112.01488, https://arxiv.org/abs/2112.01488.

- [S3] Faysse et al. (2025), ColPali, arXiv:2407.01449, https://arxiv.org/abs/2407.01449.
- [S4] Yang et al. (2025), Qwen3 Technical Report, arXiv:2505.09388, https://arxiv.org/abs/2505.09388.
- [S5] Qwen (2025), Qwen3-Reranker-4B model card, https://huggingface.co/Qwen/Qwen3-Reranker-4B.
- [S6] Microsoft GraphRAG query documentation, https://microsoft.github.io/graphrag/query/overview/.